

Difference between var, let and const

The var, let, and const keywords in JavaScript are used for variable declaration, but differ significantly in their behavior, scope, and usage. var, the oldest keyword, has function scope, meaning variables declared with var are accessible throughout the entire function they're declared in, not just within the block where they were defined.

Feature var let const

Scope Function scope Block scope Block scope

Hoisting Hoisted (initialized as undefined) Hoisted (Temporal Dead Zone) Hoisted (Temporal Dead Zone)

Reassignment ' Allowed ' Allowed 'LNot allowed

Redeclaration ' Allowed in same scope 'LNot allowed 'LNot allowed

JavaScript Data Types

JavaScript has 8 basic data types (7 primitive + 1 object type):

1. Primitive Data Types (7 types)

1.1 Number

javascript

```
let integer = 42;
```

```
let float = 3.14;
```

```
console.log("abc" / 2); // NaN
```

1.2 String

javascript

```
let singleQuote = 'Hello';
```

```
let doubleQuote = "World";
```

```
let backticks = `Template literal ${singleQuote}`;
```

```
console.log(typeof singleQuote); // "string"
```

```
console.log(`Sum: ${5 + 3}`); // "Sum: 8"
```

1.3 Boolean

```
let isTrue = true;
```

```
let isFalse = false;
```

```
console.log(typeof isTrue); // "boolean"
```

```
console.log(5 > 3); // true
```

1.4 Undefined

```
let notAssigned;
```

```
let explicitlyUndefined = undefined;
```

```
console.log(typeof notAssigned); // "undefined"
```

1.5 Null

```
let emptyValue = null;  
console.log(emptyValue); // null
```

1.6 Symbol (ES6)

```
let sym1 = Symbol("id");  
let sym2 = Symbol("id");
```

```
console.log(typeof sym1); // "symbol"  
console.log(sym1 === sym2); // false - each Symbol is unique
```

1.7 BigInt (ES2020)

```
let bigNumber = 123456789012345678901234567890n;  
let bigInt = BigInt("9007199254740991");
```

```
console.log(typeof bigNumber); // "bigint"  
console.log(9007199254740991n + 1n); // 9007199254740992n
```

2. Object Type (Non-primitive)

2.1 Object

javascript

```
let person = {  
  name: "John",  
  age: 30,  
  isStudent: false  
};
```

```
console.log(typeof person); // "object"  
console.log(person.name); // "John"
```

2.2 Array (Special type of object)

javascript

```
let fruits = ["apple", "banana", "orange"];  
console.log(typeof fruits); // "object"  
console.log(Array.isArray(fruits)); // true
```